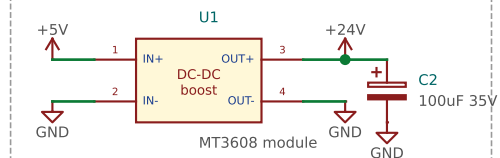
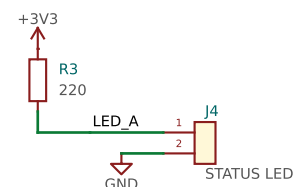
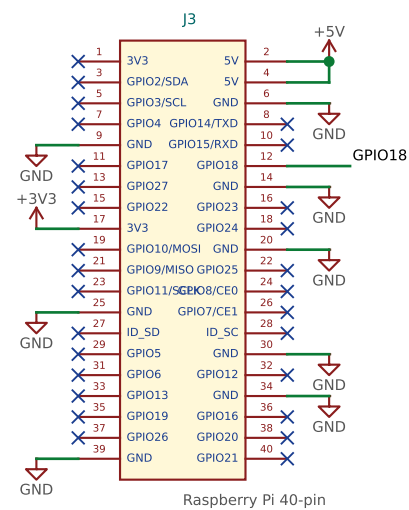


PTC fuse + TVS: a reversed plug or an adapter above ~6.5 V makes D1 conduct and F1 trip, protecting the Pi.



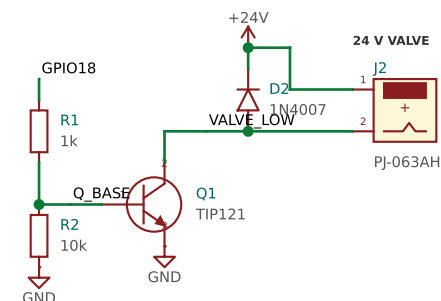
Set the module to 24.0 V with its trimmer
BEFORE fitting it (approx. 150 mA valve load).

RASPBERRY PI 40-PIN (HAT, socket on bottom side)



On while the Pi 3.3 V rail is up.
J4 pin 1 = LED +, pin 2 = LED -

VALVE DRIVER (24 V, low-side switch)



GPIO18 high = valve open. R2 keeps Q1 off while the Pi boots.
D2 absorbs the coil spike. J2 sleeve is NOT ground.

MOUNTING (Pi HAT, M2.5)



NOTES

1. Board outline follows the Raspberry Pi HAT size (65 x 56.5 mm, M2.5 holes). No ID EEPROM is fitted.
2. The Pi is powered through GPIO pins 2/4 (back-powering). On a Pi 5 add `usb_max_current_enable=1` to config.txt.
3. MT3608 modules differ slightly between sellers: match the IN+/IN-/OUT+/OUT- labels on the PCB to the module.
4. PJ-063AH jacks fit 5.5 x 2.1 mm plugs and are rated 8 A / 24 V.